

3.3.2 To set:

A poset $[A; R]$ is called toset if every pair of element of set A are comparable.

Toset is also known as linearly ordered set (or) chain.

If $[A, R]$ is a poset, then

$[A ; R^{-1}]$ is called dual of the poset.

i.e., these diagram of $[A : R^{-1}]$ can be obtained by turning the hasse diagram of $[A, R]$ upside down.

3.3.3 Hasse Diagram (Poset Diagram):

- It is graphical representation of a poset.
It is constructed as below:
 - Create vertex corresponding to every element of set A .
 - All loops & Edges implied from transitivity are not shown.
- Let $[A, \leq]$ be a poset
We say x covers y , if $x < y$ such that there doesn't exist any $z \in s$, $x < z < y$
- Thus hasse diagram shows edges b/w two elements x & y only if x covers y (or) y covers x . The set of such pairs $x < y$ is called covering relation of $(s, \leq)$.
- Thus applying reflexive transitive closure on covering relation of a poset gives the poset.
- Hasse Diagram of a toset is a chain.

3.3.4 Maximum Element:

In a poset an element is called maximal if it is not related to any other element.

3.3.5 Greatest element (or) Maximum element:

In a poset an element is called greatest if every element of the set relates to that element.

3.3.6 Minimal element:

In a poset an element is called minimal, if no other element is related to it.

3.3.7 Least element (or) Minimum Element:

In a poset least element is the one which relates to every other element of the set.

- Every finite, nonempty lattice has at least one minimal and one maximal element.
- Greatest or least element if exists is unique.
- Greatest and least elements may or may not exist if they exist they are the only maximal and only minimal elements respectively.

3.3.8 Upper Bound:

If $[A : R]$ is a poset and $B \leq A$

Upper band of $B = \{x | \forall b \in B, x \leq b \text{ and } x \in A\}$

3.3.9 Least Upper Bound (lub or join or Supremum):

In a poset $[A : R]$ lub of two elements $a, b \in A$ is least element of upper bound of $\{a, b\}$. It is denoted as $a \vee b$.

i.e., $a \vee b \leq x \forall x \in \text{upper band of } [a, b]$

- If no least element exists in the upper bound, we say lub doesn't exist.
- In other word,
If $a \vee b = c$ then
 aRc & bRc and
If $aRd \wedge bRd \Rightarrow cRd$

3.3.10 Greatest lower Bound (glb or meet or infimum)

In a poset $[A, R]$ glb of a, b is greatest element of lower bound of $\{a, b\}$. It is denoted as $a \wedge b$

- If no such element exists we say glb doesn't exist.
- Other way to define glb is
If $a \wedge b = c$, then
 $[eRa \wedge cRb] \wedge [(dRa \wedge dRb) \Rightarrow dRc]$
- For poset $[D_n ; 1]$
 $\text{Lub}(a, b) = \text{LCM}(a, b)$
 $\text{Glb}(a, b) = \text{GCD}(a, b)$
- For poset $[P(A) ; \subseteq]$, where $P(A)$ is power set of A
 $\text{lub}(x, y) = x \cup y$
 $\text{lub}(x, y) = x \cap y$
- The hasse diagram of $[P(A); \subseteq]$ forms a hypercube on, where $|A| = n$

3.3.11 Join Semi Lattice:

It is a poset in which every pair of elements has lub.

3.3.12 Meet Semi Lattice:

It is a poset in which every pair of elements has glb.

3.4 Lattice

A lattice is a poset in which every pair of elements has both glb & lub.

i.e., Lattice is both join semi lattice & meet semi lattice.

A lattice L is denoted as $(L, \wedge, \vee)$

- It is not needed that every lattice has greatest and least element.
- Every finite lattice has least and greatest elements.

3.4.1 Properties of Lattice:

- $\left. \begin{array}{l} a \wedge a = a \\ a \vee a = a \end{array} \right\}$ Idempotent
- $\left. \begin{array}{l} a \vee b = a \vee a \\ a \wedge b = a \wedge a \end{array} \right\}$ Commutative
- $\left. \begin{array}{l} a \vee (b \vee c) = (a \vee b) \vee c \\ a \wedge (b \wedge c) = (a \wedge b) \wedge c \end{array} \right\}$ Associative
- $\left. \begin{array}{l} a \vee (a \wedge b) = a \\ a \wedge (a \vee b) = a \end{array} \right\}$ Absorption law
- $a \leq b \Rightarrow a \vee c \leq b \vee c$
 $a \leq b \Rightarrow a \wedge c \leq b \wedge c$

If $a \leq b$ and $c \leq d$ then

$$a \vee c \leq b \vee d$$

$$a \wedge c \leq b \wedge d$$

3.4.2 Sublattice:

If A is a lattice B is called

Sublattice of A iff

- B itself is a lattice.
- lub of any two element of B is same as lub of the two elements in A.



In above figure B is subset of A and B is a lattice still B is not a sublattice of A.

Because $\left. \begin{array}{l} \text{in A } glb(2,3) = 4 \\ \text{in B } glb(2,3) = 5 \end{array} \right\}$ not equal $\therefore$ not sublattice

3.4.3 Types of Lattices:

(1) Bounded Lattice:

- A lattice with greatest element and least element is called bounded lattice.
- Every finite lattice is bounded lattice.

(2) Complemented Lattice:

- Complemented lattice is a bounded in which every element has atleast one complement. B is said to be complement of a iff.
- $\text{glb}(a, b) = \text{greatest element}$ and $\text{lub}(a, b) = \text{least element}$.

(3) Distributive Lattice:

- A lattice is said to be distributive if following distributive laws hold.
- $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$.
- $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$.

Note: If L is a bounded distributive lattice then complement of an element if exists, is unique. The reverse need not to be true.

Hence if we find more than one complement for an element, we can conclude that the lattice is not distributive

(4) Boolean Algebra:

- A lattice which is both distributive and complemented is known as Boolean algebra
- It is called so because it satisfies all the properties of Boolean algebra. Thus when given lattice is a Boolean algebra we can apply all the rules that we apply in logic.
- In Boolean algebra every element has exactly one complement.
- $[D_n : 1]$ is a distributive lattice for any n.
- It is because distributive properties hold for lcm & gcd.
- $[D_n : 1]$ is a Boolean lattice if n is a square free number.
- $[P(s); \leq]$ is a Boolean lattice.
- Every Boolean lattice with 2^n elements, $\forall n \geq 0$.
Every Boolean lattice contains 2^n elements is isomorphic to the lattice $(P(s); \leq)$ where s is a set with n elements.
Also this Boolean lattice is a hypercube Q_n .
- Sublattice of a complemented lattice need not to be a complemented lattice.
- Sublattice of a bounded lattice is a bounded lattice.

3.5 Function (or) Mapping (or) Transformation

A function F from set A to set B is an assignment of exactly in one element of B to each element of A. It is denoted as: $F : A \rightarrow B$

Here A is called Domain B is called codomain

Function is a special type of relation.

- Consider $f(a) = b$
b is called image of a
a is called preimage of b.